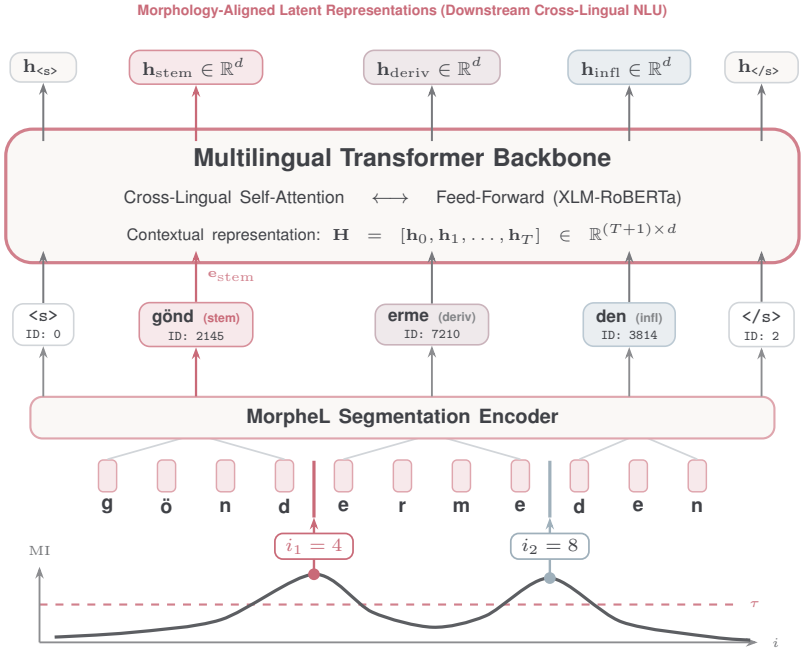


5. Pretrained LM Encoding

Feeds morphology-aligned token IDs into multilingual XLM-R backbone, generating contextual representations $\mathbf{H}$.

4. Inference & Fallback (RLM)

Fast $O(1)$ query via cache $\mathcal{C}_{\text{seg}}$; unseen OOV words fall back to on-the-fly MI and Recursive Longest Match.

3. Deterministic Cache & Vocab

Freezes canonical argmax cuts $(T = 0)$ into cache $\mathcal{C}_{\text{seg}}$ and induces vocabulary $\mathcal{V}$ with base alphabet coverage.

2. Stochastic Cut Sampling

Samples $0, 1$, or 2 cuts via Gumbel-max on cut logits γ/T for training-time morphological regularization.

1. MI Boundary Discovery

Detects candidate boundaries where $\text{MI}(w, i) > \tau$, guided by language vowel inventory $\mathcal{V}_\ell$ and harmony rules.